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ORIGINAL ARTICLE

An automated PCOS prediction model using hybrid text and gate network model using learning approaches

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Abstract
PCOS is a hormonal disorder that primarily results in female infertility in women who are of childbearing age. Increased body hair, irregular periods, increased acne, and obesity are common signs of PCOS. To control the symptoms and lower the hazards to one's health, PCOS must be detected early. The diagnosis is made using the Rotterdam criteria, which include polycystic ovaries on ultrasound imaging, ovulation failure, and a high level of androgen hormones. PCOS identification is a challenging PCOS diagnostic criterion. To manually identify PCOSs, doctors and radiologists currently employ ovarian ultrasonography to count the number of follicles and assess their volume in the ovaries. In addition to the patient's medical complaints, these healthcare professionals must do additional tests and examinations for biochemical and clinical signs in order to identify PCOS. Two adapted deep learning architectures like a feature-sequence convolutional network and a gated feature-learning network are employed as base learners to automatically extract discriminative patterns from tabular clinical variables. After that, this work created a deep learning model with the TextNet (TxNet) and GateNet (GNet) with a 97% accuracy rate in identifying the PCOS from the provided. Then, this work suggested a hybrid model that uses Clinical data to ascertain if a patient has polycystic ovary syndrome (PCOS). By utilizing the TxNet and GNet architecture to extract the clinical features and combining them with PCOS information, the best model that has been constructed attained an accuracy of 96%.

Keywords PCOS · Prediction · Deep learning · Accuracy · Text features

The Bold Texts represents the values related to the proposed methodology

PCOS Prediction Using Hybrid TxNet and GNet Models

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1 Introduction

Nowadays, the most common health problem impacting women in their reproductive years is polycystic ovarian syndrome (PCOS), which affects 5–10% of them. One endocrine condition that frequently contributes to infertility is PCOS (Hosseini et al. 2023). The failure to release the egg from the ovary is known as infertility. The first sign of PCOS is thought to be an irregular quantity and size of follicles developing throughout the ovulation stage, which is one of the factors that lead to infertility (Escobar-Morreale 2018). This fluid is found inside a woman's ovary and is stored in tiny cysts called ovarian follicles. PCOS is characterized by increased acne, alopecia, obesity, infertility, excessive facial and body hair, and irregular or non-existent periods (Teede et al. 2018). Several studies suggest that enhanced androgen production in ovarian theca cells is the underlying cause of the sickness, despite the fact that the specific cause is unknown (Barber and Franks 2021). Despite the intense controversy surrounding the diagnostic criteria for this hormonal condition, the clinical validation of PCOS is typically accepted as the standard. Three factors form the basis of the Rotterdam criteria. A diagnosis of PCOS may be made if two of these symptoms appear. The following are the three aspects: (1) oligomenorrhea, (2) polycystic ovaries (PCOM), and (3) elevated levels of androgen (male sex hormone). For early PCOS prediction, one of the most important tools is ovarian ultrasound imaging. Important details on the quantity, size, and location of the follicles are included in this image of the ovary (Wo´ zniak et al. 2021). The most popular imaging method for clinically assessing ovarian cancer patients is ultrasound. Compared to other medical imaging techniques like computed tomography (CT) and magnetic resonance imaging (MRI), ultrasound offers a number of benefits. Ultrasonography's affordability, safety, and immediate findings lend credence to this. Apart from providing an excellent opportunity to develop a deep learning model for independent examination, this imaging technique improves the test's objectivity and diagnostic accuracy (Saleem and Rizvi 2017). Deep learning is a strong technique in computer vision and analysis where early detection of PCOS using automatic classification using ultrasound images and clinical information aids in identifying the condition.

Ultrasound exams, blood tests, and superior monthly data are necessary for the PCOS diagnosis, which is determined by a variety of factors and symptoms. Considering the vast range of symptoms associated with this condition and the absence of a single diagnostic procedure or approach that physicians use to assess patients doctors recommend inconsistent clinical test findings and unneeded radiological imaging (Gheisari et al. 2024). By excluding symptoms or test results, PCOS is identified as a condition that is not connected to the condition because its intricate pathomechanism is poorly understood. At the same time, since PCOS directly causes ovarian dysfunction, early identification and detection with the fewest possible Imaging techniques and laboratory testing are crucial. As such, it increases the chance of gynecological cancer, abortion, and infertility. Additionally, patients experience mental distress as a result of the time and money wasted (Bhosale et al. 2022). Currently, a manual diagnosis is made for polycystic ovarian morphology in ultrasound imaging. Experts' knowledge and expertise in identifying the features and makeup of ovarian ultrasound pictures serve as its foundation. After looking at the same case's photos, the expert's judgment may occasionally be arbitrary and changeable. Additionally, compared to less experienced physicians, when done by experts, ultrasound diagnosis accuracy is better. But there aren't many specialized examiners (Maadi et al. 2021), particularly in less developed areas. Additionally, a radiologist manually estimates the sizes and numbers of follicles on the ovary images. However, because follicles vary in size and are associated with veins and tissues, radiologists find it difficult and time-consuming to detect PCOM. Additionally, it introduces noise and artifacts into the photographs (Hu et al. 2019). Long-term effects on women's fertility and hormone imbalance may result from diagnostic ambiguity. Additionally, this manual diagnosis framework may lead to more inaccuracies in the examination, which would be inconvenient for the patient. However, it is advised that gynecologists use intelligent computer programs that can give them tools for making decisions—applying an algorithm for deep learning to evaluate a woman's ultrasound images and medical records to ascertain if she is suffering from PCOS. Additionally, it will assist in overcoming the challenges of manually examining ultrasound images and evaluating patient clinical data.

Recently, a range of tasks, such as detection, segmentation, and classification, have been applied to medical ultrasound imaging using deep learning and machine learning (Tiwari et al. 2022). In CAD, ultrasound scans of different body parts have been utilized to diagnose a variety of life-threatening conditions, including prostate cancer, hydronephrosis and breast cancer. Furthermore, other researchers have made numerous contributions to the identification of PCOS by the use of ultrasound imaging (Nasim et al. 2022). Several deep learning and models for machine learning, including CNN, NB, SVM, and VGG-16 have been applied to the interpretation of ovarian ultrasound images for diagnostic systems. Instead of using ultrasound images to diagnose PCOS, other studies have used different data types, such as text-formatted ultrasonography results (Danaei Mehr and Polat 2022) and clinical data. Additionally, as shown in (Maheswari et al. 2021), researchers discovered that CNN designs such as Google LeNet, VGG-16, and U-Net have produced noteworthy outcomes for a range of computer vision applications, such as segmentation and classification of photos. To expedite training and improve performance, a deep learning technique was used by doing away with the necessity of manually extracting features from the images. During model training, the features can be automatically extracted by CNN. Furthermore, it has been noted that earlier studies built their suggested model using a private dataset. Although the study employed an open-source dataset, it included 3D images that are inappropriate for our investigation. Furthermore, most of these studies built their models using a small-sized dataset. Generally speaking, there is a dearth of PCOS diagnoses made with ultrasound scans. Furthermore, based on clinical and metabolic data, we found that machine-learning algorithms have been employed to diagnose PCOS patients in (Nandipati et al. 2020). Nevertheless, polycystic ovaries, ovulation failure, and elevated androgen hormones on ultrasound imaging (PCOM) are all part of the Rotterdam criteria for PCOS diagnosis. Few previous studies have suggested a deep learning-based fusion model that can investigate how clinical information and ultrasound pictures interact to diagnose PCOS. The use of images and health data using a "fusion model" to address complex issues that are not be answered by a single modality, such as Alzheimer's disease, breast cancer and skin cancer has increased in the literature on modern medical imaging. Therefore, the model still needs to be improved to diagnose PCOS more accurately, utilizing clinical data and ovarian ultrasound images. The development of the suggested model to identify PCOS is aided by the two studies on applied computational intelligence and soft computing mentioned. This study's major purpose is to establish a PCOS computer-aided diagnostic (CAD) model that will assist radiologists in classifying images from ovarian ultrasounds. Although ultrasound imaging plays an important role in clinical PCOS diagnosis, the present study does not utilize ultrasound images as model inputs. Instead, ultrasound-related indicators are implicitly reflected through structured clinical and biochemical variables contained in the dataset. The proposed framework focuses exclusively on tabular clinical data, aiming to provide a lightweight and accessible PCOS prediction tool without reliance on imaging modalities. The objectives are to increase the model's accuracy and decrease the false positive rate. To help the physician and radiologist make better clinical decisions, investigate the influence of clinical factors on using ultrasound pictures and a deep learning fusion method to diagnose PCOS. The following are this work's primary contributions:

1. A highly imbalanced dataset is pre-processed using various techniques to prepare it for deep learning models with excellent performance.
2. Models for automatic feature extraction using a modified TextNet (TxNet) and GateNet (GNet) are suggested as an alternative to manually constructed approaches.
3. Three distinct, straightforward, yet powerful deep learning models are developed, used, and evaluated in order to increase the accuracy of PCOS predictions.
4. Several current PCOS detection techniques are compared with this version to illustrate how much better the proposed DL model is.

The following is the structure of the paper: Sect. 2 provides a more thorough examination of the several widely used methods. The technique is drafted in Sect. 3. In Sect. 4, the results are displayed numerically. Section 5 provides the conclusion.

2 Related works

The examination of pertinent earlier research papers and commonly used state-of-the-art research on PCOS prediction is covered in this section. These papers were selected due to their pertinence to the subject matter. One of the most common medical conditions identified in younger women with a range of symptoms is PCOS (Jeevitha and Priya 2022). Early PCOS detection is crucial to preventing long-term problems. Experts employed machine learning techniques to achieve this goal. As recommended, the random forest approach yielded a 96% accuracy score in diagnosing PCOS. A new machine learning-based method known as CS-PCOS was suggested to choose the Thale dataset's most noteworthy characteristic for PCOS detection based on the optimal chi-squared algorithm. While waiting for help to become accessible, women can use the online application provided by Reference (Khanna et al. 2023) to quickly and easily determine their risk of getting the disease from the comfort of their homes. Among the machine learning algorithms that the authors worked with, the best results were produced by the KNN classifier. To reduce the subset of characteristics and concentrate on the most pertinent ones, the authors combined the machine learning-based methods discussed with feature selection techniques (AlamSuha and Islam 2023). Various methods yield excellent and efficient results for these fine-tuned features, including naïve Bayes, random forest, and multi-layer perceptrons. A new diagnostic paradigm for PCOS was developed by employing RNA-seq and microarray datasets, using methods from artificial neural networks and random forests (Wang et al. 2023), which have shown a greater capacity for generalization with microarray data. Similarly, an approach that combined the random forest prediction model with the feature selection strategy was also published.

Blood serum profiles from women with and without PCOS were gathered as part of a Raman spectroscopy study. Apparent variations in brightness between individuals with PCOS and those without the disorder were found in the collected blood profiles. Based on the data, the ratio of proteins to lipids may serve as a helpful sign of PCOS that can be detected by Raman spectroscopy. Researchers are employing deep learning techniques in addition to machine learning-based approaches by using CNNs to identify relevant characteristics in ultrasound image datasets (Pandey et al. 2023). Furthermore, a deep learning technique was used to predict PCOS from ultrasound pictures. The "SCBOD" count-based ovarian detection model was developed. The initial stage of this model, which has four stages, is cleaning the ultrasound visual data. The second stage uses the segmentation technique to pick and detect items. Using a range of metrics, including dimension, weight, average, variance, and others, SCBOD correctly identifies items in the third stage using its analytical and architectural features (Elreedy and Atiya 2019). An SVM is utilized for classification to identify PCOS in a patient. Additionally, the scientists in extracted geometrical features from ultrasound pictures using multi-level thresholding. The retrieved features are fed into a supervised learning system to determine whether an image has PCOS. QLattices, ELI5, LIME, and other explainable artificial intelligence (Zigarelli et al. 2022) techniques have recently been coupled with a multi-stack machine learning model to identify PCOS. The author in (Denny et al. 2019) employed an hybrid-based framework that extracted the most salient features from a dataset and developed and evaluated five conventional machine learning models to obtain the most accurate prediction. This involved several techniques, including PCA and Chi-Square. While machine learning-based approaches were also incorporated (Faris and Miften 2022), tongue and pulse readings were used to predict PCOS. On the other hand, the authors of another study used machine learning to identify that Diabetes, high blood pressure, and other cardiovascular conditions are among the signs of PCOS (Alshakrani et al. 2022). Researchers created a new dataset for this study by combining two pre-existing public datasets. Following feature selection, the eight selected features have been linked to the disease using supervised and unsupervised techniques.

Convolutional neural networks (CNNs) outperform other neural networks on visual input. Therefore, based on visual inputs, the IAKmeans-RSA technique (Saeed 2023) has been proposed for use in follicle recognition and cyst segmentation. A CNN was used to extract the relevant information from the segmented images. The classification is done in the last phase utilizing a deep neural network (DNN) method (Nicolucci et al. 2022). Much work has been done using visual examples and studies based on textual data, such as pulse readings. Hybrid deep learning-based algorithms have been utilized to predict PCOS based on ultrasound images. Ultrasound pictures have been subjected to an hybrid model based on deep learning to predict PCOS. This study presented a hybrid CNN that assessed the probability of PCOS using VGG-16 and ResNet-50 settings that have already been trained (Rajalakshmi et al. 2024). An MRI image was used as the input for developing the ResNet-50 and pre-trained VGG-16 architectures. The VGG-16 model's final max pooling layer's result was then sent to a layer with a ReLu activation function enabled that was fully connected (FC) (Dileep et al. 2023). The FC layer, which had a ReLu installed, simultaneously receives the output of the ResNet-50's final average pooling layer.

3 Materials and methods

This section will outline the suggested method for predicting PCOS. The proposed deep learning-based models are described in detail after a quick overview of the dataset is provided in this section.

3.1 Dataset description

Kottarathil supplied the data collection on the diagnosis of PCOS utilized in this work, which is accessible through the Kaggle repository. 541 individuals' comprehensive and diagnostic data are included in the dataset. The study offers a file containing clinical and physical traits linked to PCOS, primarily focusing on screening and diagnosis. Two of the data's 45 factors have been found to have unique identifying values, and one of them has been eliminated since it contains NULL data. The goal variable "PCOS" is binary since a 1 denotes a positive case and a 0 denotes a negative case. A final count of 540 samples remained after the dataset was cleaned up by removing non-numerical information and an inconsistent patient record. We resample our data using SMOTE to guarantee a balanced distribution of losses across the training and testing datasets. Testing uses 20% of the dataset, while training uses 80%. Figure 1 shows the flow diagram of the proposed model.

3.2 Pre-processing

A correlation analysis is carried out to enhance the PCOS diagnosis dataset's quality and utility. The objective of this analysis is to find columns that have weak correlations with other dataset variables. As a result, a correlation analysis between the dataset's various variables is carried out using the Pearson correlation coefficient. Columns with a correlation value less than 0.1 were removed because they didn't significantly impact the data. All columns with a correlation of 0.1 or greater and up to 1.0 are considered. Such columns were eliminated, leaving 35 columns in the dataset. This process is frequently employed in feature selection. Remarkably, cutting out features unrelated to the dataset makes it simpler to model accurately using the target variable or other features. The conservative correlation value of 0.1 excludes only features with a weak link. As a result, the dataset is smaller, and diagnosing PCOS without infertility may become more accurate. The original dataset exhibited an imbalanced class distribution between PCOS-positive and PCOS-negative samples, which can bias the learning process toward the majority class and reduce sensitivity for minority-class detection. To mitigate this, we applied the SMOTE exclusively to the training partition after performing the train–test split, thereby preventing information leakage into the test set. The minority class was oversampled to achieve a balanced class ratio (1:1) within the training data. The test set was kept unchanged and reflects the original class distribution to ensure an unbiased evaluation.

3.3 Prediction

Combining the predictions of several NNs is a hybrid TxNet and GNet strategy in DL that enhances overall performance. Due to its ability to transcend the limits of individual models, it offers benefits, including decreased variance

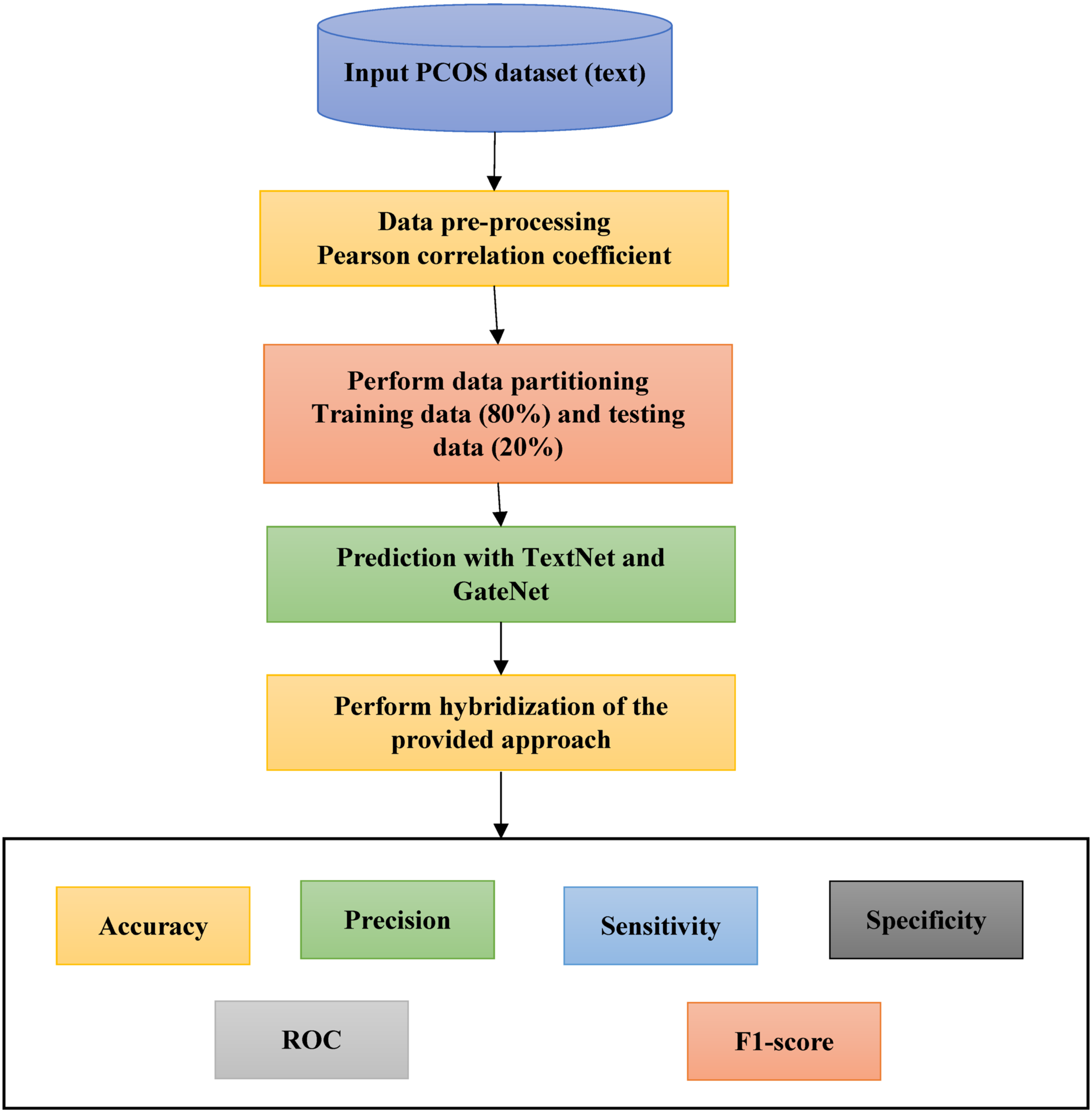


Fig. 1 Flow diagram of the proposed model

and generalization error. These methods are employed in numerous industries, including healthcare and finance, such as image identification, intelligent grid theft, and heart disease detection. The models are trained in hybrid TxNet and GNet techniques, and their predictions are aggregated using various procedures such as voting, stacking, and averaging. The following are some of the most widely used hybrid techniques: TxNet and GNet. Figure 2 illustrates the two hybrid strategies proposed in this study that are blended as basis models in proposed model which uses NN as the meta-learner layer.

3.4 Hybrid model

In this work, TextNet (TxNet) and GateNet (GNet) are adapted deep learning architectures rather than entirely new network designs. Both networks are modified and customized to process one-dimensional structured clinical data for PCOS prediction. The architectural adaptations include the

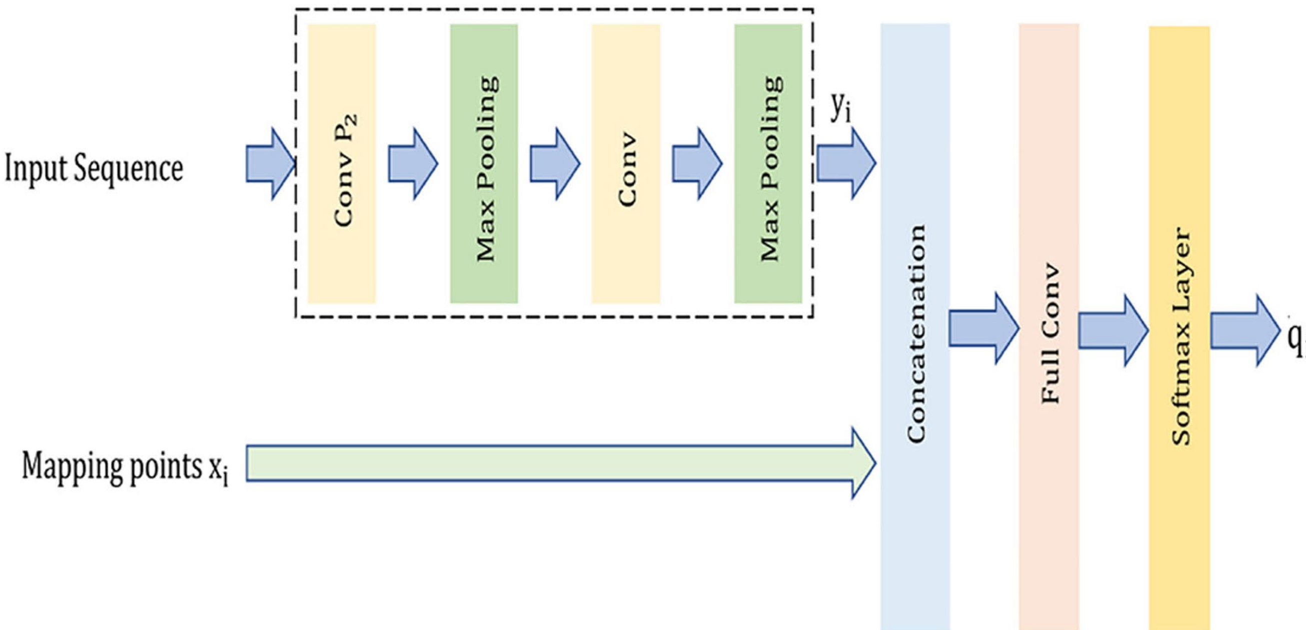


Fig. 2 TxNet architecture

use of one-dimensional convolutional layers, task-specific kernel sizes, and tailored activation functions to suit the characteristics of clinical feature vectors. A hybrid TxNet and GNet model creates a single predictive model by combining several DL models. It maximizes each model's positive aspects while minimizing its negative ones. The hybrid TxNet and GNet model outperforms a single model for prediction because it combines several models. The two methods that can be employed to generate a hybrid TxNet and GNet model are sequential hybrid modelling and parallel hybrid TxNet and GNet modelling. Sequential hybrid models generate a single output by feeding the second base model with the first base model's production. In a parallel hybrid TxNet and GNet model, each model runs independently. The predictions from both models are based on the same input. After that, the outputs of both models are combined to form a single production, thus becoming a hybrid model. Numerous elements, including the hyper-parameter values and the training procedures must be carefully considered while developing a hybrid TxNet and GNet model. The two DL models that make up our suggested model. When both models are used together, the performance is improved. The input for both models was the same. The network processes the input using fully connected, max-pooling, and convolutional layers. The output is derived from features by appropriately varying the convolutional layers' strides and kernel sizes. The proposed PCOS model does not use temporal sequencing; all inputs are structured clinical features processed as fixed-length vectors. The final hybrid TxNet and GNet layer receives the concatenated output from both models and applies it to a final forecast. In Algorithm 1, the workflow is displayed. The proposed hybrid architecture integrates two base learners like TxNet and GNet are trained on the same clinical input features. The probability outputs (or penultimate-layer embedding's) from TxNet and GNet are concatenated and provided to an MLP meta-learner to produce the final PCOS prediction. The exact number of trainable parameters and layer-wise configuration depend on the selected input feature size after pre-processing and the hyper-parameter settings; therefore, the final architecture statistics reported in this work correspond specifically to the implemented PCOS configuration.

3.5 Learning models for PCOS

A final forecast is produced by combining the results from several models using the blending process. This lowers the likelihood that the model will become over fit and improves the blending-based model's overall performance. TxNet and GNet, two foundation models, plus MLP, a meta-model, have been combined in this study to create TGNet. Training, validation, and testing are the three dataset sections separated for mixing. The base models like TxNet and GNet are trained on the training set. After training, predictions are made using the validation set. These forecasts serve as the new characteristics used to train the MLP meta-model. MLP gains knowledge of these novel properties and correctly produces the final prediction on unknown data.

3.6 Network model

One of the first CNN designs includes the seven layers that comprise architecture include three convolutional, two pooling, and two fully connected. As the model's name implies, five of these seven levels are learnable. Figure 2 depicts network architecture. TxNet is inspired by one of the earliest convolutional neural network architectures originally developed for handwritten digit recognition. In this study, TxNet is adapted into a one-dimensional convolutional architecture to process structured clinical data rather than two-dimensional image inputs. This adaptation enables the model to capture local dependencies and feature interactions among adjacent clinical variables, which are critical for PCOS prediction. For this reason, the network is employed in this study to detect PCOS problems. Convolutional layers are essential because they help uncover hidden patterns and complex correlations in the data. By reducing the feature maps' dimensionality without compromising the pooling layers, which contain the most important information, which come after the convolutional layers simplify the network. The first convolutional layer uses five kernels and thirty-two neurons to perform a 1D convolution on 1D PCOS data. A pooling layer comes next. Using the same kernel size and eight neurons, the second convolutional layer improves the feature maps even more. With 16 neurons, the third convolutional layer uses the 1×1 convolution. Then come three fully connected layers containing 32, 16, and 4 neurons. The final dense layer employs a sigmoid activation function, while the first five learnable layers use a tanh activation function. The likelihood of having heart disease is output by this last layer. This combination of activation functions is used to maintain a balance between computing efficiency and network non-linearity. The network is a straightforward model that works well. This makes TxNet a well-liked option in the DL space. For challenges involving the prediction of PCOS, alternating convolutional and pooling layers in the TxNet architecture, followed by fully linked layers, offers a strong foundation. Figure 3 shows the GNet model.

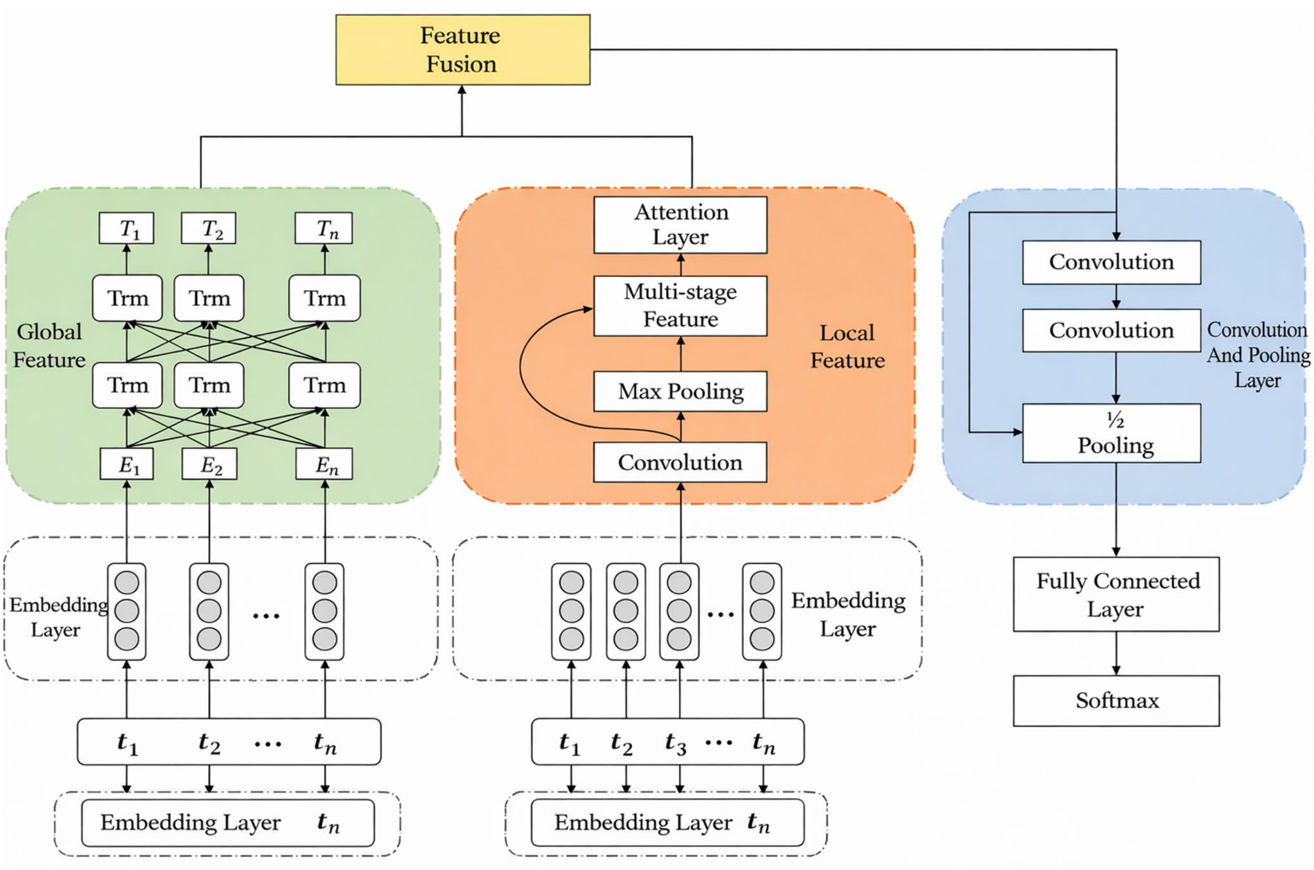


Fig. 3 GNet model

Input: PCOS dataset $D = \{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$

1: Perform data partition with training data D_{train}, testing data D_{test};

2: Input text sequence of PCOS data;

3: Initialize TxNet and GNet with hyper-parameters and network architecture;

4: Train TxNet on D_{train} by providing input PCOS sequence to TxNet;

5: Acquire TxNet based prediction on D_{train};

6: Train GNet model with D_{train} by providing PCOS input X via with GNet model;

7: Acquire GNet based prediction on D_{train};

8: Hybridize both TxNet and GNet model with valid outcomes;

9:Train the forward MLP with valid outcomes on X_{test};

10: Finalize PCOS based on the prediction

Fig. 4 PCOS Prediction Using Hybrid TxNet and GNet Models ...

4 Performance analysis

This section displays the research' findings and provides a description of the various metrics, including recall, accuracy, and precision. Accuracy, recall, precision, F1-score, and AUC are among the metrics used to evaluate the performance of the suggested model. The following is the definition of these metrics. The following is how accuracy is expressed:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

The high proportion of true positives among all positive outcomes is shown by the recall parameter.

$$Recall = \frac{TP}{TP + FN} \tag{2}$$

Recall, sometimes called sensitivity or actual positive rate, offers information on how well the algorithm can identify pertinent data. True positives divided by all positives is the definition of precision.

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

Another crucial performance indicator is the F1-score, which is the harmonic mean of precision and recall.

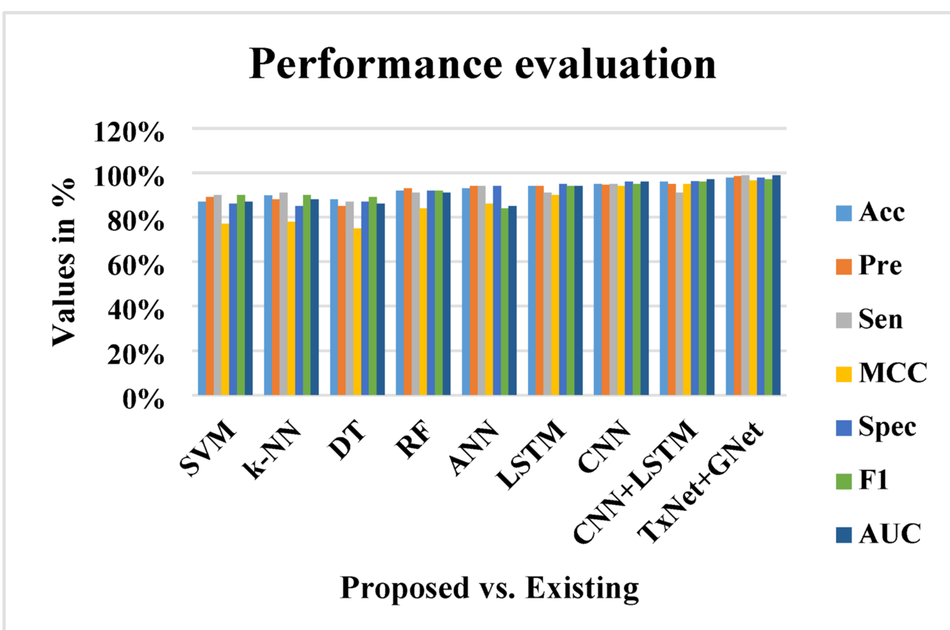
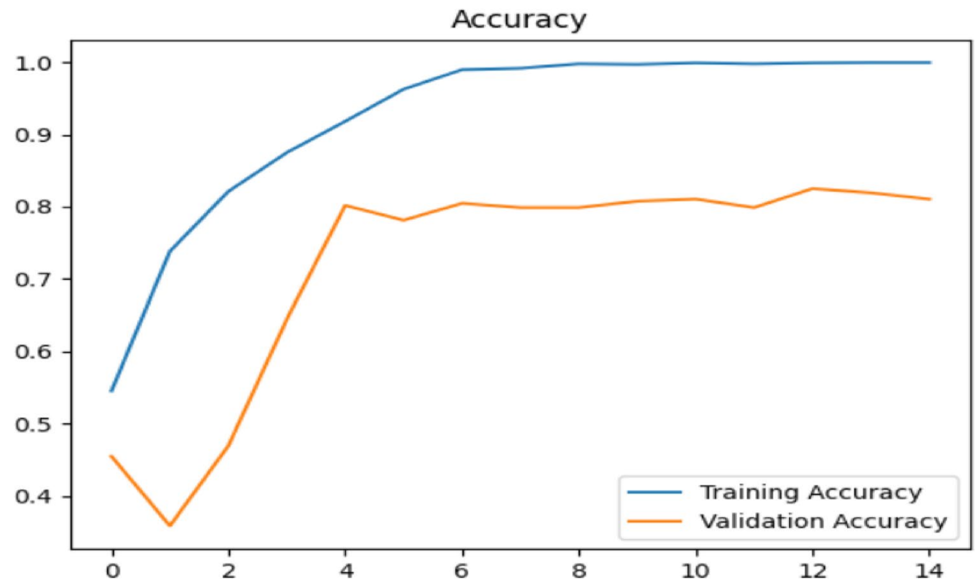


Fig. 5 Comparative analysis

Fig. 6 Accuracy comparison



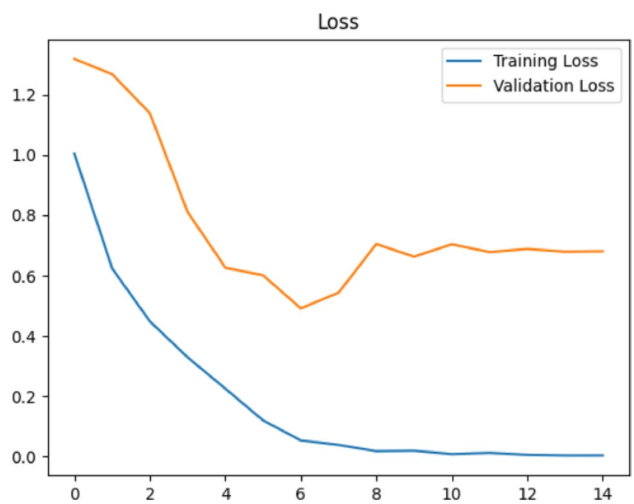


Fig. 7 Loss comparison

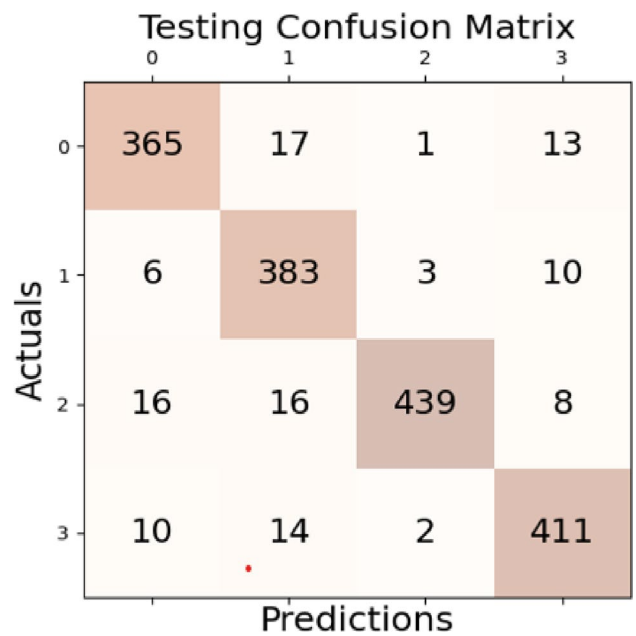


Fig. 8 Confusion matrix

Table 1 Optimal parameters

Parameters	Values
Filters	8
Kernel size	4
Activation function	Tanh
Loss	Binary
Learning rate	0.01
Optimizer	Adam
Epochs	50
Batch	32
No. of neurons	1

$$F1 - score = 2 * \frac{Precision * Recall}{Precision + Recall} \tag{4}$$

The area under the curve (AUC) indicates the degree of discriminability or separation. It demonstrates how well our model distinguishes between the various classifications. Better classification performance is indicated by a larger AUC value, which precisely separates the patient and healthy classes. Fig. 5, 6, 7.

4.1 Simulation outcomes

Different tuning parameters were used in the test, and the outcomes were recorded for each configuration. The final configurations for the ideal parameters discovered for adjusting the models were as follows: Adam was the optimizer; binary cross-entropy was the loss function; A learning rate of 0.01 and an epoch count of 50 were established.

Batches of 32 and so on were subjected to gradient descent. Table 1 has a list of them.

The performance outcomes for the specified environment are noted and displayed in Table 1. The table shows testing for a number of traits, including the AUC, F1-score, recall, accuracy, and precision. Additionally, confusion matrices and graphs are provided to analyze the parameters. The concept of the true positives discovered is given by the independent presentation of the confusion matrices for every model. The test and train data are also shown against the number of epochs to display accuracy and loss. Figure 8 shows the confusion matrices for the suggested models. The three suggested PCOS prediction models have equivalent accuracy measures of 92, 96, and 94%. The percentage of patients accurately identified out of all patients acquired is known as recall in our approach. Models one, two, and three had peak recalls of 92, 96, and 94%, respectively where the percentage of individuals with PCOS correctly recognized out of all those recognized as patients. The three models have respective precision scores of 93, 96, and 94%. A summary of the relevant data points is provided by precision, which helps assess the model's accuracy. The model yields the most outstanding results, as seen in Fig. 5. This model has the highest precision, recall, F1-score, and AUC scores, according to the confusion matrix. The hybrid TxNet and GNet model exhibit the best performance for all specified parameters.

Feature extraction is done using a one-dimensional TGNet. Although it performs marginally worse than the prior stack TGNet is still somewhat effective. The least amount of convergence is seen in the TxNet and GNet model. CNNs are inherently spatial feature-extracting bodies which explains why they outperformed the other two models. Alternately, temporal relationships can be detected by TxNet. The TxNet and GNet produces good results even if it adds temporal reliance on top of spatial dependency. The CNN becomes even more complicated as a result. As a result, the CNN performs better than the TxNet and GNet. The plots in Fig. 6 depict the models' gradual epoch-specific convergence. The accuracy vs. epoch charts demonstrate that the TxNet and GNet model performs well; the variance of the model is low. This model's gap between the training

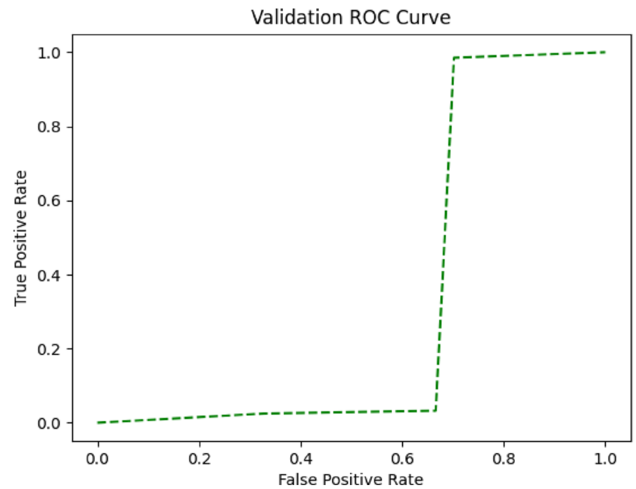


Fig. 9 ROC comparison

and testing curves is lower. The training and testing data for the other models differ noticeably. With 96.59% accuracy, the TxNet and GNet model had the highest score, whereas the accuracy scores of the other two models were considerably lower.

Furthermore, Fig. 7 indicate the loss experienced versus epochs. While the testing loss exhibits only mild convergence, the TxNet and GNet model's training loss abruptly converges. The TxNet and GNet model displays the best convergence with the least bias and volatility among all the models. In this model, the loss of the test set is equal to the loss of the training set. When the TxNet and GNet is exposed to the training and testing sets, it exhibits discernible variance. Figure 9 illustrates the expected algorithm's AUC behavior. Model-1 had an AUC of 92.0%, Model-2 got an AUC of 96.60%, and Model-3 got an AUC of 94.3%. All three models do well in terms of this goal, as seen by the graphs that provide an overview of their AUCs. Once more, the TxNet and GNet model achieves a top score of 96.59%, outperforming all the others. Therefore, based on the findings and further analysis, Model 2, which combines SMOTE with CNN, outperforms the other two models. Compared to earlier research, our model is very effective, particularly because it is lightweight and can be trained with

Table 2 Performance comparison of PCOS prediction models

Model	Acc	Pre	Sen	MCC	Spec	F1	AUC
SVM	87%	89%	90%	77%	86%	90%	87%
k-NN	89.8%	88%	91%	78%	85%	90%	88%
DT	88%	85%	87%	75%	87%	89%	86%
RF	92%	93%	91%	84%	92%	92%	91%
ANN	93%	94%	94%	86%	94%	84%	85%
LSTM	94%	94%	91%	90%	95%	94%	94%
CNN	95%	94.6%	95%	94%	96%	95%	96%
CNN+LSTM	96%	95%	91%	95%	96.2%	96%	97%
TxNet+GNet	**97.8%**	**98.5%**	**98.8%**	**96.6%**	**97.8%**	**97.1%**	**98.8%**

fewer parameters. The CNN model uses only 297 parameters to train, but the TxNet and GNet models need 13,285 parameters. There are 6689 parameters in the LSTM model. The suggested model saves time and parameters; it finished training in 10.02 s, far quicker than the CNN + LSTM combination model (18.51 s) and the LSTM model (67.27 s). The efficiency measures, such as the number of trainable parameters, training time, and computational resources used by each model is displayed in Table 2. To assess whether the observed differences in ROC–AUC between models are statistically significant, we applied DeLong's test for two correlated ROC curves, which is appropriate because the AUCs were computed on the same test set (paired predictions). Pairwise comparisons were conducted between the proposed model (TxNet+GNet / TGNet) and each baseline model (CNN, LSTM, and CNN+LSTM). A two-sided significance level of $\alpha = 0.05$ was used. The resulting p-values indicate whether the AUC improvement of the proposed model over each baseline is statistically significant.

4.2 Comparative analysis

To determine how effective the suggested paradigm is, the performance of the same dataset must be fairly compared. The purpose of this section is to provide a comparative examination of a few recent and contemporary PCOS detection models. The capacity of several models to diagnose PCOS is displayed in Table 2. The same Kaggle dataset was used to obtain these models and the previously mentioned standard performance metrics were used to compare the models' performances. An innovative diagnosis method for PCOS called TGNet was created by evaluating the effectiveness of numerous machine learning classifiers. The existing system uses random forest (RF) and sqrt as the maximal feature hyper-parameter has demonstrated the best accuracy of 93. Some authors recently looked into various machine learning classifiers to find a good way to detect PCOS. They used the linear discriminant analysis (LDA) technique to attain a precision of 96% and an accuracy of 92%. The authors examined several machine learning techniques to address the issue of precise PCOS identification. Methods like bagging, Adaboost, naïve Bayes, KNN, SVM, RF, and neural networks were investigated. With 40 features, the random forest classifier produced the best results (accuracy = 93.12%, precision = 93.12%, and recall = 93.12%). The self-diagnostic model for PCOS proposed had a precision of 95% and an accuracy of 90% across various analysis scenarios. In contrast, a different collection of hybrid TxNet and GNet classifiers was examined including CatBoost, voting soft, and voting hard. According to the inquiry and analysis, using voting software to predict patients with PCOS yielded a maximum accuracy of 91.12%. The author investigated different classifiers using the model to forecast PCOS. Figure 9 shows the ROC comparison.

According to their analysis, random forest and data sampling produced the best classification performance, with 95% accuracy, 96% precision, and 94% recall. Recently, an SVM approach based on genetic algorithms was proposed to categorize PCOS. The performance outcomes were not particularly compelling. A study that used various machine learning classifiers for PCOS detection and prediction was also published. The precision was 95%, and the accuracy was 89%. The authors proposed an approach founded on a combination of linear regression and random forest (RFLR) with 91% accuracy, 97% precision, 92% recall, and 92% area under the curve. Many hybrid models, such as linear SVM, light gradient boosting, extreme boosting using RF, and CatBoost, were examined to detect PCOS. According to the assessments, CatBoost outperformed all other hybrid TxNet and GNet models with a 92% accuracy rate, a 95% recall rate, and an 84% precision rate. Our suggested model built on a lightweight 1D-CNN is a substantially superior predictor of PCOS with a significantly higher accuracy rate of 96%, 96% precision rate, 96% recall rate, and 96% F1-score and AUC of 96%. Table 2 summarizes the comparative analysis based on performance indicators, including AUC, F1-score, accuracy, precision, and recall. The suggested deep learning model outperforms all of the PCOS models regarding overall prediction performance. Figure 9 also shows the ROC comparison. Therefore, compared to several current predictive and detecting models for PCOS, the suggested model performs better, as the comparative analysis confirms.

4.3 Discussion

We made a bias-invariant prediction in our investigation using three models. The final predictor, TxNet and GNet is selected as the model that yields the best results. The TxNet is utilized on the training set of the dataset after it has been divided into training and test parts, while the test set remains unchanged. Deep learning performs exceptionally effectively with scaled and normalized data, which is why the data are likewise normalized and scaled. Various preparation processes are applied to the dataset, such as eliminating NULL features and using the Pearson correlation feature selection technique. With a peak accuracy of 96%, the model shows impressive convergence on the datasets used for training and testing. Other significant outcomes from the finished model are also shown in these diagrams. As a result, there was more convergence in the model with a TGNet feature extractor added. The experiment findings demonstrated that using a TGNet had a noticeable effect on this dataset. After a few epochs, the loss smoothed out despite our best efforts to minimize it. Because of this, we stopped our model early to avoid over-fitting. For later usage, the finished model is saved.

5 Conclusion

Numerous deep learning techniques and machine learning classifiers have been investigated for the diagnosis and prognosis of PCOS. Handcrafted feature extraction utilizing machine learning-based algorithms has been frequently recommended in practice for the accurate identification and prognosis of PCOS; however, these methods have low-performance issues that can be ignored. This study suggested using lightweight deep learning models in conjunction with automated feature engineering to improve the accuracy of PCOS diagnosis. Instead of using machine learning techniques for feature extraction, which require several data pre-processing steps to prepare a highly unbalanced dataset for high performance and validity, the proposed PCOS prediction approach provides deep learning models based on TxNet and customized GNet. This will enhance the effectiveness and quality of the medical systems used to diagnose PCOS. To improve the accuracy of PCOS predictions, three distinct yet effective lightweight deep learning models are developed and analyzed. In terms of accuracy, precision, recall, F1-score, and ROC-AUC, our hybrid TxNet and GNet model outperformed the other recommended models, achieving scores of 97.6, 97, and 97.5%, respectively. With training completion times of 10 s, the suggested model is incredibly time-efficient. It is also more efficient than the other two models because it uses the fewest parameters and the results of the test which compares the AUCs of the predicted models demonstrate its statistical importance and usefulness. The recommended model provides the best performance, which could help with early PCOS detection. Furthermore, the suggested model performs better than several newly developed current PCOS prediction and detection learning algorithms

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Declarations

Conflict of interest There are no conflicts of interest to declare.

Consent to participate This article does not contain any studies involving animals performed and any studies involving human participants performed by any of the authors.

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